

Name a number that is a multiple of both 5 and 10.

Number: _____

True or false?

18 is a multiple of both 4 and 6.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 1

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 10.

LCM: _____

Find the least common multiple of 8 and 12.

LCM: _____

What is the least common multiple of 4 and 9?

A. 72

B. 13

C. 18

D. 36

Find the least common multiple of 5 and 15.

LCM: _____

True or false?

The least common multiple of 6 and 9 is 54.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{2}{3} + \frac{1}{4}$$

Value: _____

Find the value.

$$\frac{3}{5} + \left(-\frac{7}{10}\right)$$

Value: _____

Find the value.

$$-\frac{1}{2} + \left(-\frac{3}{8}\right)$$

A. $-\frac{7}{8}$

B. $-\frac{4}{10}$

C. $\frac{1}{8}$

D. $-\frac{1}{8}$

Find the value.

$$-1\frac{1}{3} + \frac{5}{6}$$

Value: _____

True or false?

$$-\frac{3}{4} + \frac{3}{4} = 0$$

A. True

B. False

Subtract fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Subtracting fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 24 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

Subtract.

$$\frac{7}{8} - \frac{1}{2}$$

Difference: _____

Subtract.

$$\frac{4}{5} - \frac{2}{3}$$

A. $\frac{2}{2}$

B. $\frac{1}{5}$

C. $\frac{6}{15}$

D. $\frac{2}{15}$

Subtract.

$$\frac{9}{10} - \frac{2}{5}$$

Difference: _____

Subtract.

$$1 - \frac{3}{7}$$

Difference: _____

SET 24 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{3}$?

A. $\frac{10}{15}$

B. $\frac{4}{9}$

C. $\frac{3}{4}$

D. $\frac{2}{6}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{3}{4} = \frac{9}{16}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 2

Subtracting fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Subtract.

$$\frac{7}{9} - \frac{2}{9}$$

Difference: _____

True or false?

$$\frac{5}{8} - \frac{3}{8} = \frac{2}{8}$$

A. True

B. False

$$\frac{11}{12} - \frac{4}{12} = ?$$

A. $\frac{7}{24}$

B. $\frac{7}{12}$

C. $\frac{15}{12}$

D. $\frac{4}{12}$

Subtract.

$$1 - \frac{3}{5}$$

Difference: _____

True or false?

$$\frac{9}{10} - \frac{4}{10} = \frac{5}{20}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 6.

Multiples: _____

True or false?

15 is a multiple of both 3 and 5.

A. True

B. False

Which number is a multiple of both 4 and 5?

- A.** 15
- B.** 20
- C.** 9
- D.** 25

Name a number that is a multiple of both 3 and 8.

Number: _____

True or false?

16 is a multiple of both 4 and 6.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{3} = \frac{5}{6}$$

x = _____

Solve for y.

$$y - \frac{2}{5} = \frac{3}{10}$$

y = _____

Solve for n.

$$\frac{3}{4} + n = 1\frac{1}{2}$$

A. $1\frac{1}{4}$

B. $\frac{3}{4}$

C. $2\frac{1}{4}$

D. $\frac{1}{4}$

Solve for m.

$$m + \frac{5}{8} = 2$$

m = _____

True or false?

The solution of $x - \frac{1}{6} = \frac{2}{3}$ is $x = \frac{5}{6}$.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$\frac{1}{4} - \frac{5}{6}$$

Value: _____

Find the value.

$$-\frac{2}{3} - \frac{1}{6}$$

Value: _____

Find the value.

$$\frac{1}{2} - \left(-\frac{3}{4}\right)$$

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. $1\frac{1}{4}$

D. $-1\frac{1}{4}$

Find the value.

$$-\frac{3}{5} - (-\frac{1}{2})$$

Value: _____

True or false?

$$\frac{2}{9} - \frac{5}{9} = \frac{1}{3}$$

A. True

B. False

Add and subtract mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Adding and subtracting mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Mixed numbers and fractions greater than 1	4.NF.B.3.b
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Solve equations of the form $px + q = r$ with fractions	7.EE.B.4.a

SET 25 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$2\frac{1}{3} + 1\frac{1}{2}$$

Sum: _____

Subtract.

$$5\frac{3}{4} - 2\frac{1}{6}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{1}{2}$$

A. $3\frac{3}{10}$

B. $2\frac{3}{10}$

C. $3\frac{7}{10}$

D. $2\frac{7}{10}$

Add.

$$3\frac{5}{8} + 2\frac{3}{4}$$

Sum: _____

Subtract.

$$6 - 2\frac{2}{3}$$

Difference: _____

SET 25 • 5.NF.A.1

BACKWARD 1

Adding and subtracting mixed numbers with like denominators • Grade 4 •
Number & Operations–Fractions • 4.NF.B.3.c

Add.

$$1\frac{2}{5} + 2\frac{1}{5}$$

Sum: _____

Subtract.

$$4\frac{5}{6} - 1\frac{4}{6}$$

Difference: _____

True or false?

$$2\frac{3}{4} + 1\frac{3}{4} = 3\frac{6}{8}$$

A. True

B. False

$$5\frac{1}{3} - 2\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $2\frac{1}{3}$

C. $2\frac{2}{3}$

D. $3\frac{2}{3}$

Add.

$$3\frac{7}{8} + \frac{3}{8}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 2

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{17}{5}$ as a mixed number.

Mixed number: _____

True or false?

$$1\frac{1}{3} = \frac{4}{3}$$

A. True

B. False

Which fraction is equal to $3\frac{1}{2}$?

A. $\frac{31}{2}$

B. $\frac{7}{2}$

C. $\frac{5}{2}$

D. $\frac{6}{2}$

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

SET 25 • 5.NF.A.1

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{2}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{1}{2}$?

A. $\frac{5}{10}$

B. $\frac{2}{3}$

C. $\frac{2}{2}$

D. $\frac{1}{4}$

Find the missing number.

$$\frac{2}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{5}{6} = \frac{10}{18}$$

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 2\frac{1}{2} = 6\frac{1}{4}$$

x = _____

Solve for y.

$$y - 1\frac{2}{3} = 3\frac{1}{6}$$

$$y = \underline{\hspace{10cm}}$$

Solve for n.

$$n + 4\frac{3}{8} = 7$$

A. $3\frac{5}{8}$

B. $2\frac{3}{8}$

C. $11\frac{3}{8}$

D. $2\frac{5}{8}$

A board was $8\frac{1}{2}$ feet long. After a piece was cut off, it was $5\frac{3}{4}$ feet long.

Write and solve an equation to find the length p of the piece.

Equation and solution: _____

True or false?

The solution of $x - 1\frac{1}{4} = 2\frac{1}{2}$ is $x = 1\frac{1}{4}$.

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 2

Solve equations of the form $px + q = r$ with fractions • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$2x + \frac{1}{2} = 3\frac{1}{2}$$

x = _____

Solve for x.

$$\frac{1}{3}x - 1\frac{1}{4} = 2\frac{3}{4}$$

x = _____

Solve for x.

$$4x - 2\frac{1}{3} = 1$$

A. $1\frac{1}{12}$

B. $\frac{1}{6}$

C. $\frac{5}{6}$

D. $\frac{1}{3}$

Jo bought 3 bags of flour that weigh the same and one $1\frac{1}{2}$ -pound bag of sugar. The total weight was $9\frac{3}{4}$ pounds.

How much does each bag of flour weigh?

Weight: _____

True or false?

The solution of $\frac{1}{2}(x + 3) = 4$ is $x = 5$.

A. True

B. False

Solve word problems by adding and subtracting fractions

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Word problems with like denominators	4.NF.B.3.d
BACKWARD 2	Choosing addition or subtraction	2.OA.A.1
BACKWARD 3	Adding and subtracting fractions with unlike denominators	5.NF.A.1
FORWARD 1	Write and solve equations of the form $x + p = q$	6.EE.B.7
FORWARD 2	Write and solve equations of the form $px + q = r$	7.EE.B.4.a

SET 26 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Mia walked $\frac{3}{4}$ mile to school and then $\frac{2}{3}$ mile to the park.

How far did she walk in all?

Distance: _____

A jug had $\frac{7}{8}$ gallon of milk. Ben drank $\frac{1}{3}$ gallon.

How much milk is left?

Milk left: _____

Ana used $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar.

How many cups did she use in all?

A. $4\frac{1}{4}$ cups

B. $3\frac{3}{4}$ cups

C. $4\frac{3}{4}$ cups

D. $3\frac{1}{4}$ cups

A ribbon is $5\frac{1}{6}$ feet long. Lea cuts off $2\frac{1}{2}$ feet.

How long is the ribbon now?

Length: _____

Tom read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of the book on Tuesday.

What fraction of the book does he still have to read?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 1

Word problems with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

Kim ate $\frac{3}{8}$ of a pizza and Joe ate $\frac{2}{8}$ of the pizza.

What fraction of the pizza did they eat in all?

Fraction: _____

A rope is $\frac{9}{10}$ meter long. $\frac{4}{10}$ meter is cut off.

How much rope is left?

Rope left: _____

True or false?

$\frac{2}{6}$ of a garden is corn and $\frac{3}{6}$ is beans. Corn and beans cover $\frac{5}{6}$ of the garden.

A. True

B. False

Lu had $\frac{7}{12}$ of a cake. She gave away $\frac{5}{12}$ of the cake.

How much of the cake is left?

A. $\frac{12}{12}$

B. $\frac{2}{12}$

C. $\frac{35}{12}$

D. $\frac{2}{24}$

A tank is $\frac{3}{5}$ full. Water is added to fill another $\frac{1}{5}$ of the tank.

How full is the tank now?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 2

Choosing addition or subtraction • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

A class has 14 boys and 13 girls.

Which operation finds the number of students in all?

- A.** Subtraction
- B.** Division
- C.** Addition
- D.** Multiplication

True or false?

To find how much is left after some is used, you subtract.

- A.** True
- B.** False

Sam had 42 marbles. He lost 15 marbles.
How many marbles are left?

Marbles: _____

A shelf holds 35 books. 18 books are on it now.

Which equation finds how many more books fit?

A. $35 - 18 = \square$

B. $18 - 35 = \square$

C. $35 \times 18 = \square$

D. $35 + 18 = \square$

Liz has 26 stickers. She gets 17 more.

How many stickers does she have now?

Stickers: _____

SET 26 • 5.NF.A.2

BACKWARD 3

Adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{3} + \frac{1}{2}$$

Sum: _____

Subtract.

$$\frac{3}{4} - \frac{1}{3}$$

Difference: _____

True or false?

$$\frac{1}{2} + \frac{1}{4} = \frac{2}{6}$$

A. True

B. False

$$\frac{5}{6} - \frac{1}{2} = ?$$

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{4}{4}$

D. $\frac{1}{6}$

Add.

$$\frac{2}{5} + \frac{1}{2}$$

Sum: _____

SET 26 • 5.NF.A.2

FORWARD 1

Write and solve equations of the form $x + p = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

After using $\frac{2}{3}$ cup of oil, Ray has $\frac{3}{4}$ cup left.

Write and solve an equation to find how much oil c he started with.

Equation and solution: _____

A plant grew $\frac{5}{8}$ inch and is now $4\frac{1}{4}$ inches tall.

Write and solve an equation to find its starting height h .

Equation and solution: _____

$\frac{3}{10}$ inch of rain fell on day 1. In all, $\frac{9}{10}$ inch fell on days 1 and 2.

Which equation finds the rain r on day 2?

A. $\frac{3}{10}r = \frac{9}{10}$

B. $r + \frac{9}{10} = \frac{3}{10}$

C. $\frac{3}{10} + r = \frac{9}{10}$

D. $r - \frac{3}{10} = \frac{9}{10}$

Solve for x.

$$x + \frac{2}{9} = \frac{2}{3}$$

x = _____

True or false?

If $n - \frac{1}{4} = \frac{5}{8}$, then $n = \frac{7}{8}$.

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 2

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A recipe uses $\frac{3}{4}$ cup of sugar per batch, plus $\frac{1}{2}$ cup for frosting. Ali used $2\frac{3}{4}$ cups of sugar in all.

How many batches did he make?

Batches: _____

Solve for x.

$$\frac{2}{3}x + \frac{1}{6} = \frac{5}{6}$$

x = _____

A 6-foot board is cut into 4 equal pieces, with $\frac{1}{2}$ foot left over.

How long is each piece?

A. $1\frac{1}{2}$ feet

B. $1\frac{5}{8}$ feet

C. $1\frac{3}{8}$ feet

D. $5\frac{1}{2}$ feet

Solve for x.

$$5x - \frac{3}{4} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $2x + \frac{1}{3} = 1$ is $x = \frac{2}{3}$.

A. True

B. False

Use benchmark fractions to estimate and to check answers

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Comparing fractions to $\frac{1}{2}$ and 1	4.NF.A.2
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Reason about the size of a quotient of fractions	6.NS.A.1
FORWARD 2	Use estimation to judge whether an answer is reasonable	7.EE.B.3

SET 27 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

$\frac{2}{5} + \frac{1}{2} = \frac{3}{7}$ must be wrong, because $\frac{3}{7}$ is less than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{12}$?

- A.** About $\frac{1}{2}$
- B.** About $2\frac{1}{2}$
- C.** About $1\frac{1}{2}$
- D.** About 3

Is $\frac{3}{5} + \frac{4}{9}$ more or less than 1?

Use benchmark fractions to explain.

Answer: _____

Explanation: _____

Jen ran $\frac{9}{10}$ mile and walked $\frac{1}{8}$ mile.

Estimate the total distance to the nearest whole mile.

Estimate: _____

Leo says $\frac{5}{6} - \frac{1}{3} = \frac{4}{3}$.

Which statement explains why his answer is not reasonable?

- A.** The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.
- B.** The denominators are different.
- C.** $\frac{4}{3}$ is less than 1.
- D.** The answer should be 0.

SET 27 • 5.NF.A.2

BACKWARD 1

Comparing fractions to $\frac{1}{2}$ and 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{5}{8}$ is less than $\frac{1}{2}$.

A. True

B. False

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{3}{8}$

C. $\frac{2}{5}$

D. $\frac{4}{7}$

Is $\frac{5}{9}$ closest to 0, $\frac{1}{2}$, or 1?

Answer: _____

True or false?

$\frac{6}{10}$ is less than $\frac{1}{2}$.

A. True

B. False

Is $\frac{11}{12}$ closer to $\frac{1}{2}$ or to 1?

Answer: _____

SET 27 • 5.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

Which point is at $\frac{1}{3}$?



A. Point A

B. Point B

C. Point C

D. Point D

What fraction is at point B?



B = _____

True or false?

$\frac{7}{8}$ is closer to 1 than to $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 1

Reason about the size of a quotient of fractions • Grade 6 • The Number System • 6.NS.A.1

True or false?

$\frac{3}{4} \div \frac{1}{8}$ is greater than 1.

A. True

B. False

Without dividing, which quotient is less than 1?

A. $\frac{3}{4} \div \frac{1}{2}$

B. $\frac{1}{3} \div \frac{1}{2}$

C. $2 \div \frac{1}{2}$

D. $\frac{5}{6} \div \frac{1}{3}$

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

Quotient: _____

Estimate $2\frac{7}{8} \div \frac{1}{2}$ by first rounding $2\frac{7}{8}$ to the nearest whole number.

Estimate: _____

True or false?

$\frac{1}{2} \div 4$ is greater than $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A board $9\frac{7}{8}$ feet long is cut into 3 equal pieces.

Estimate the length of each piece.

Estimate: _____

Ann earns \$11.85 per hour and works $7\frac{3}{4}$ hours.

Which is the best estimate of her pay?

- A.** \$96
- B.** \$84
- C.** \$120
- D.** \$77

True or false?

$\frac{3}{4}$ of \$79.95 is about \$60.

A. True

B. False

Kai says $4\frac{5}{6} + 3\frac{1}{8}$ is about 9.

Is this reasonable? Give a better estimate.

Reasonable? _____

Better estimate: _____

A recipe needs $2\frac{1}{3}$ cups of flour per batch.

Estimate the flour needed for 4 batches.

Estimate: _____

Interpret a fraction as division of the numerator by the denominator

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	Division with remainders	4.NBT.B.6
BACKWARD 3	Mixed numbers and fractions greater than 1	4.NF.B.3.b
FORWARD 1	Unit rates written as fractions	6.RP.A.2
FORWARD 2	Write fractions as decimals using division	7.NS.A.2.d

SET 28 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 5$ as a fraction.

Fraction: _____

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{4}$?

A. $7 \div 4$

B. $7 - 4$

C. $4 \div 7$

D. 7×4

Write $\frac{13}{6}$ as a division expression. Then write the quotient as a mixed number.

Division: _____

Mixed number: _____

True or false?

When 2 pounds of rice are shared equally among 3 bags, each bag gets $\frac{2}{3}$ pound.

- A. True
- B. False

SET 28 • 5.NF.B.3

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ is one of 4 equal parts of a whole.

A. True

B. False

A sandwich is cut into 3 equal parts.

What fraction of the sandwich is one part?

Fraction: _____

What fraction of the bar is shaded?



A. $\frac{5}{3}$

B. $\frac{3}{5}$

C. $\frac{2}{5}$

D. $\frac{3}{2}$

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 2

Division with remainders • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide. Write the quotient and the remainder.

$$17 \div 5$$

Quotient and remainder: _____

True or false?

$$23 \div 4 = 6 \text{ R } 1$$

A. True

B. False

$$29 \div 6 = ?$$

A. 5 R 5

B. 5 R 1

C. 4 R 3

D. 4 R 5

Divide. Write the quotient and the remainder.

$$50 \div 8$$

Quotient and remainder: _____

True or false?

$$15 \div 2 = 7 \text{ R } 2$$

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 3

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{9}{4}$ as a mixed number.

Mixed number: _____

Write $1\frac{2}{3}$ as a fraction.

Fraction: _____

True or false?

$$\frac{11}{5} = 1\frac{1}{5}$$

A. True

B. False

Which mixed number is equal to $\frac{7}{3}$?

A. $2\frac{2}{3}$

B. $3\frac{1}{3}$

C. $2\frac{1}{3}$

D. $1\frac{1}{3}$

Write $\frac{15}{4}$ as a mixed number.

Mixed number: _____

SET 28 • 5.NF.B.3

FORWARD 1

Unit rates written as fractions • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A recipe uses 3 cups of flour for every 4 cups of water.
How many cups of flour are used per cup of water?

Cups of flour: _____

8 apples cost \$6.

What is the cost per apple?

Cost per apple: _____

A car goes 150 miles on 4 gallons of gas.
What is the rate in miles per gallon?

- A.** $\frac{4}{150}$
- B.** 600
- C.** 154
- D.** 37.5

There are 5 cups of juice for every 2 pitchers.
How many cups of juice are there per pitcher?

Cups per pitcher: _____

True or false?

The ratio 3 : 8 has the unit rate $\frac{3}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

FORWARD 2

Write fractions as decimals using division • Grade 7 • The Number System •
7.NS.A.2.d

Write $5 \div 8$ as a decimal.

Decimal: _____

Use long division to write $\frac{7}{12}$ as a decimal.

Decimal: _____

Which fraction has a decimal form that repeats?

A. $\frac{7}{20}$

B. $\frac{5}{6}$

C. $\frac{9}{25}$

D. $\frac{3}{8}$

Write $11 \div 4$ as a decimal.

Decimal: _____

True or false?

$$\frac{1}{3} = 0.3$$

A. True

B. False

Solve sharing problems whose answers are fractions or mixed numbers

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	Whole numbers as fractions	3.NF.A.3.c
FORWARD 1	Solve unit rate problems	6.RP.A.3.b
FORWARD 2	Unit rates with fractional quantities	7.RP.A.1

SET 29 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

6 friends share 10 granola bars equally.

How many bars does each friend get? Write a mixed number.

Bars: _____

A 50-pound bag of rice is shared equally among 8 families.

How many pounds does each family get?

Pounds: _____

9 yards of ribbon are cut into 4 equal pieces.

How long is each piece?

A. $4\frac{1}{2}$ yards

B. $\frac{4}{9}$ yard

C. $2\frac{1}{9}$ yards

D. $2\frac{1}{4}$ yards

7 liters of juice are poured equally into 3 jugs.

How many liters are in each jug? Between which two whole numbers is your answer?

Liters: _____

Between: _____

True or false?

When 5 cookies are shared equally by 2 children, each child gets $2\frac{1}{2}$ cookies.

A. True

B. False

SET 29 • 5.NF.B.3

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

24 stickers are shared equally by 6 friends.

How many stickers does each friend get?

Stickers: _____

35 chairs are set out in 5 equal rows.

How many chairs are in each row?

A. 30

B. 6

C. 40

D. 7

True or false?

If 4 boxes hold 28 toys, with the same number in each box, each box holds 7 toys.

A. True

B. False

A baker puts 45 muffins into bags of 9.
How many bags does she fill?

Bags: _____

18 pencils are shared equally by 3 students.

How many pencils does each student get?

Pencils: _____

SET 29 • 5.NF.B.3

BACKWARD 2

Whole numbers as fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

True or false?

$$\frac{6}{3} = 3$$

A. True

B. False

Write 4 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 3?

A. $\frac{3}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{12}$

D. $\frac{12}{4}$

How many halves are in 5 wholes?

Halves: _____

True or false?

$$\frac{8}{8} = 8$$

A. True

B. False

SET 29 • 5.NF.B.3

FORWARD 1

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

3 pounds of grapes cost \$7.

What is the cost per pound?

Cost per pound: _____

A car travels 100 miles in 3 hours at a constant speed.
At this rate, how far does it travel in 6 hours?

Distance: _____

8 cans of soup cost \$6.

At this rate, how much do 12 cans cost?

A. \$16

B. \$8

C. \$9

D. \$4

A runner runs 5 miles in 40 minutes at a constant pace.
At this pace, how long does she take to run 8 miles?

Time: _____

True or false?

If 4 pens cost \$3, then 10 pens cost \$7.00.

- A. True
- B. False

SET 29 • 5.NF.B.3

FORWARD 2

Unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail crawls $\frac{2}{3}$ foot in $\frac{1}{4}$ hour.

What is its speed in feet per hour?

Speed: _____

A recipe uses $\frac{3}{4}$ cup of sugar for every $\frac{1}{2}$ cup of milk.

How many cups of sugar are used per cup of milk?

Cups of sugar: _____

Jo walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour.

What is her speed?

A. 2 miles per hour

B. 3 miles per hour

C. $\frac{1}{3}$ mile per hour

D. $\frac{1}{12}$ mile per hour

A pump fills $\frac{5}{8}$ of a tank in $\frac{1}{2}$ hour.

How much of the tank does it fill per hour?

Tanks per hour: _____

True or false?

$\frac{1}{2}$ cup of juice for every $\frac{1}{4}$ cup of water is a rate of 2 cups of juice per cup of water.

- A.** True
- B.** False

Multiply a fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A fraction times a whole number as repeated unit fractions	4.NF.B.4.b
BACKWARD 2	Parts of a whole	3.NF.A.1
BACKWARD 3	Splitting a number into equal parts	3.OA.A.2
FORWARD 1	Find a percent of a quantity	6.RP.A.3.c
FORWARD 2	Multistep percent problems	7.RP.A.3

SET 30 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times 12$$

Product: _____

Multiply.

$$5 \times \frac{2}{3}$$

Product: _____

What is $\frac{2}{5}$ of 20?

A. 10

B. 8

C. 4

D. 50

Multiply.

$$\frac{5}{8} \times 6$$

Product: _____

True or false?

$\frac{1}{3} \times 15$ means 15 split into 3 equal parts, with 1 of the parts taken.

- A.** True
- B.** False

SET 30 • 5.NF.B.4.a

BACKWARD 1

A fraction times a whole number as repeated unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Write $\frac{4}{5}$ as a whole number times $\frac{1}{5}$.

Expression: _____

True or false?

$$3 \times \frac{2}{7} = \frac{6}{21}$$

A. True

B. False

$$3 \times \frac{3}{8} = ?$$

A. $3\frac{3}{8}$

B. $\frac{6}{8}$

C. $\frac{9}{8}$

D. $\frac{9}{24}$

Multiply.

$$4 \times \frac{2}{9}$$

Product: _____

True or false?

$$2 \times \frac{3}{5} = \frac{6}{10}$$

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 2

Parts of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ of a bar means 1 of 4 equal parts.

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

What fraction is 2 parts of size $\frac{1}{3}$?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{3}{2}$

How many $\frac{1}{8}$ s are in $\frac{5}{8}$?

Answer: _____

True or false?

$\frac{3}{5}$ means 3 parts of size $\frac{1}{5}$.

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 3

Splitting a number into equal parts • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.2

12 is split into 4 equal parts.

How much is in each part?

Answer: _____

True or false?

20 split into 5 equal parts gives 4 in each part.

A. True

B. False

Which expression shows 18 split into 3 equal parts?

A. $3 \div 18$

B. $18 - 3$

C. $18 \div 3$

D. 18×3

Divide.

$$24 \div 8$$

Quotient: _____

30 is split into 6 equal parts.

How much is in each part?

Answer: _____

SET 30 • 5.NF.B.4.a

FORWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 48?

Answer: _____

What is 60% of 35?

Answer: _____

A \$40 game is 15% off.

How many dollars are taken off the price?

- A.** \$6
- B.** \$15
- C.** \$4
- D.** \$34

A class has 30 students. 70% of them ride the bus.
How many students ride the bus?

Students: _____

True or false?

50% of 18 is 36.

A. True

B. False

SET 30 • 5.NF.B.4.a

FORWARD 2

Multistep percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A \$60 coat is 30% off.

What is the sale price?

Sale price: _____

A meal costs \$25. A 20% tip is added.

What is the total cost?

Total: _____

A \$80 bike has a 6% sales tax added.

What is the total cost?

A. \$86.00

B. \$4.80

C. \$85.60

D. \$84.80

A town of 500 people grew by 12%.

What is the new population?

Population: _____

True or false?

A 25% markup on a \$40 item makes the price \$50.

- A.** True
- B.** False

Multiply a fraction by a fraction

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A whole number times a fraction	4.NF.B.4.b
BACKWARD 2	Equivalent fractions and simplest form	4.NF.A.1
FORWARD 1	Divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 31 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{4}{5}$$

Product: _____

Multiply.

$$\frac{3}{4} \times \frac{1}{6}$$

Product: _____

Multiply.

$$\frac{5}{6} \times \frac{3}{10}$$

A. $\frac{15}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{16}$

What is $\frac{1}{2}$ of $\frac{3}{8}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 1

A whole number times a fraction • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$2 \times \frac{3}{10}$$

Product: _____

True or false?

$$5 \times \frac{1}{4} = \frac{5}{20}$$

A. True

B. False

$$4 \times \frac{2}{3} = ?$$

A. $\frac{8}{12}$

B. $\frac{8}{3}$

C. $\frac{6}{3}$

D. $\frac{2}{12}$

Multiply.

$$3 \times \frac{5}{12}$$

Product: _____

True or false?

$$6 \times \frac{1}{6} = 1$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 2

Equivalent fractions and simplest form • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{6}{8}$ as an equivalent fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{10}{15} = \frac{3}{4}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{12}$?

A. $\frac{4}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{1}{3}$

Find the missing number.

$$\frac{9}{12} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{5}{20} = \frac{1}{5}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 1

Divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{4}{5}$$

Quotient: _____

Divide.

$$\frac{3}{4} \div \frac{1}{6}$$

Quotient: _____

Divide.

$$\frac{5}{8} \div \frac{1}{4}$$

A. $\frac{5}{32}$

B. $\frac{1}{2}$

C. $2\frac{1}{2}$

D. $1\frac{1}{4}$

How many $\frac{3}{8}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-\frac{2}{3} \times \frac{3}{4}$$

Product: _____

Find the product.

$$\left(-\frac{4}{5}\right) \times \left(-\frac{5}{6}\right)$$

Product: _____

Find the product.

$$\frac{3}{7} \times \left(-\frac{7}{9}\right)$$

A. $-\frac{3}{7}$

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. $\frac{10}{16}$

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right)$$

Product: _____

True or false?

$$\left(-\frac{2}{5}\right) \times \frac{5}{2} = 1$$

A. True

B. False

Find the area of a rectangle with fractional side lengths

MAIN

Grade 5 main question

5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares

3.MD.C.7.a

BACKWARD 2

The area formula for rectangles

4.MD.A.3

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Area of triangles and other polygons with fractional measures

6.G.A.1

FORWARD 2

Area of composite figures

7.G.B.6

SET 32 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

Find the area of the rectangle.

Area: _____



A rectangle is $2\frac{1}{2}$ inches long and $\frac{4}{5}$ inch wide.

What is its area?

Area: _____

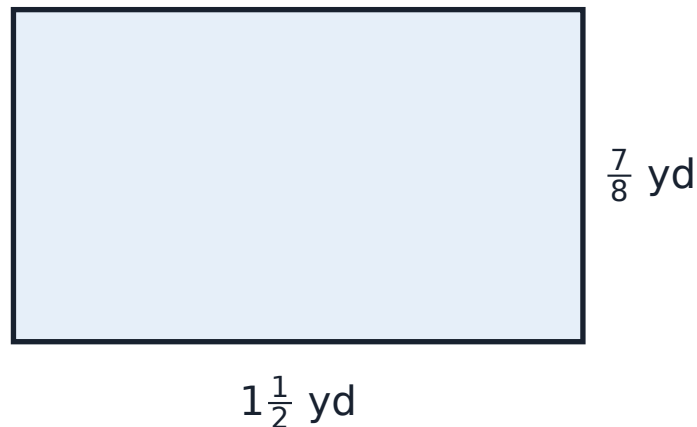
A rectangle that is $\frac{2}{3}$ unit by $\frac{3}{4}$ unit is tiled with small rectangles that are each $\frac{1}{3}$ unit by $\frac{1}{4}$ unit.

What is the area of the large rectangle?

- A.** $\frac{6}{12}$ square unit
- B.** $\frac{6}{7}$ square unit
- C.** $\frac{5}{7}$ square unit
- D.** $1\frac{5}{12}$ square units

Find the area of the rectangle.

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{2}$ foot has an area of 1 square foot.

- A.** True
- B.** False

SET 32 • 5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares, with no gaps.

What is its area?

Area: _____

True or false?

Tiling a 3-by-5 rectangle takes 8 unit squares.

A. True

B. False

How many unit squares cover a 7-by-2 rectangle?

A. 72

B. 9

C. 14

D. 18

A rectangle has 5 rows of unit squares with 8 squares in each row.

What is its area?

Area: _____

True or false?

The number of unit squares that cover a rectangle equals its length times its width.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 2

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 5 m wide.

What is its area?

Area: _____

A rectangle has an area of 54 square feet and a length of 9 feet.

What is its width?

Width: _____

True or false?

The formula $A = l \times w$ gives the area of a rectangle.

A. True

B. False

A rectangle is 15 cm long and 4 cm wide.

What is its area?

- A.** 60 square cm
- B.** 45 square cm
- C.** 38 square cm
- D.** 19 square cm

Find the area of the rectangle.

Area: _____



SET 32 • 5.NF.B.4.b

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{5}{6} \times \frac{3}{5}$$

Product: _____

Multiply.

$$\frac{5}{2} \times \frac{4}{5}$$

Product: _____

True or false?

$$\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$$

A. True

B. False

$$\frac{3}{2} \times \frac{7}{8} = ?$$

A. $\frac{10}{10}$

B. $\frac{21}{16}$

C. $\frac{21}{8}$

D. $\frac{3}{7}$

Multiply.

$$\frac{1}{2} \times \frac{1}{2}$$

Product: _____

SET 32 • 5.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures • Grade 6 •
Geometry • 6.G.A.1

A triangle has a base of $4\frac{1}{2}$ inches and a height of 2 inches.

What is its area?

Area: _____

A parallelogram has a base of $3\frac{1}{4}$ feet and a height of 4 feet.

What is its area?

Area: _____

A triangle has a base of $\frac{3}{4}$ m and a height of $\frac{2}{3}$ m.

What is its area?

A. $\frac{1}{2}$ square meter

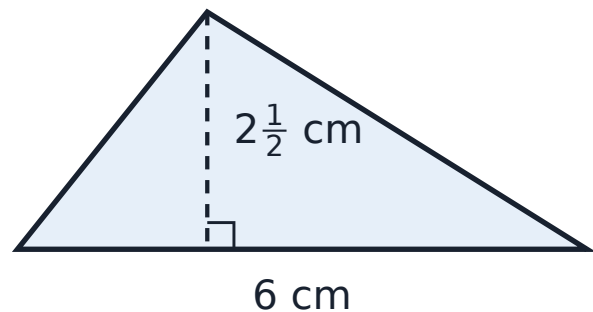
B. $\frac{5}{7}$ square meter

C. $\frac{1}{4}$ square meter

D. $\frac{6}{7}$ square meter

Find the area of the triangle.

Area: _____



True or false?

A trapezoid with bases of $2\frac{1}{2}$ m and $3\frac{1}{2}$ m and a height of 2 m has an area of 6 square meters.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

FORWARD 2

Area of composite figures • Grade 7 • Geometry • 7.G.B.6

A figure is made of a rectangle that is $6\frac{1}{2}$ ft by 4 ft and a triangle with a base of 4 ft and a height of 3 ft.

What is the total area?

Area: _____

A rectangular garden is 10 m by $7\frac{1}{2}$ m. A square patio with $2\frac{1}{2}$ -m sides is inside the garden.

What is the area of the garden that is not patio?

Area: _____

A $\frac{1}{2}$ -by-2 rectangle is cut out of a 5-by-4 rectangle.

What is the area of the part that is left?

- A.** 21 square units
- B.** 18 square units
- C.** 19 square units
- D.** 20 square units

A kite is made of two triangles. Each triangle has a base of 8 in. and a height of $3\frac{1}{2}$ in.

What is the area of the kite?

Area: _____

True or false?

A figure made of two $\frac{1}{2}$ -by-3 rectangles that do not overlap has an area of 6 square units.

- A. True
- B. False

Compare the size of a product to the size of one factor without multiplying

MAIN	Grade 5 main question	5.NF.B.5.a
BACKWARD 1	Multiplicative comparison	4.OA.A.2
BACKWARD 2	Comparing fractions to 1	4.NF.A.2
FORWARD 1	Scale quantities in ratio tables	6.RP.A.3.a
FORWARD 2	Use a scale to find lengths in scale drawings	7.G.A.1

SET 33 • 5.NF.B.5.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.a

Without multiplying, how does $\frac{3}{4} \times 20$ compare to 20?

- A.** It is less than 20.
- B.** It is equal to 20.
- C.** It is 4 times 20.
- D.** It is greater than 20.

True or false?

$5 \times 3\frac{1}{2}$ is greater than 5.

A. True

B. False

Without multiplying, is $\frac{2}{3} \times 15$ greater than, less than, or equal to 15?

Answer: _____

Which product is greater than 8?

A. $\frac{9}{8} \times 8$

B. $\frac{1}{2} \times 8$

C. $\frac{7}{8} \times 8$

D. $\frac{8}{8} \times 8$

A photo is 6 inches wide. It is resized to $\frac{5}{4}$ of its width.

Is the new width greater than or less than 6 inches?

Answer: _____

SET 33 • 5.NF.B.5.a

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.2

True or false?

3×12 is 3 more than 12.

A. True

B. False

A dog weighs 4 times as much as a 6-pound cat.
How much does the dog weigh?

Weight: _____

Which number is 5 times as much as 7?

A. 57

B. 12

C. 2

D. 35

True or false?

2×9 is less than 9.

A. True

B. False

Fill in the blank.

42 is ____ times as much as 7.

Blank: _____

SET 33 • 5.NF.B.5.a

BACKWARD 2

Comparing fractions to 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{4}{5}$ is greater than 1.

A. True

B. False

Which fraction is less than 1?

A. $\frac{6}{7}$

B. $\frac{9}{9}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

Write $>$, $<$, or $=$ to compare.

$$\frac{8}{8} \text{ ____ } 1$$

Symbol: _____

True or false?

$$\frac{3}{5} > 1$$

A. True

B. False

Is $\frac{11}{10}$ greater than, less than, or equal to 1?

Answer: _____

SET 33 • 5.NF.B.5.a

FORWARD 1

Scale quantities in ratio tables • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

A recipe uses 2 cups of rice for every 3 cups of water.
How many cups of water are needed for 8 cups of rice?

Cups of water: _____

A ratio table starts with 4 : 10.

Which ratio could also be in the table?

- A.** 4 : 5
- B.** 2 : 5
- C.** 8 : 15
- D.** 10 : 4

The table shows equivalent ratios.

Find the missing value.

Red	3	6	?
Blue	5	10	25

Missing value: _____

True or false?

Multiplying both numbers in $6 : 9$ by $\frac{1}{3}$ gives $2 : 3$, an equivalent ratio.

- A.** True
- B.** False

A map shows 5 km for every 2 cm.

How many kilometers do 7 cm show?

Kilometers: _____

SET 33 • 5.NF.B.5.a

FORWARD 2

Use a scale to find lengths in scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 inch = 4 feet. A wall is $3\frac{1}{2}$ inches long in the drawing.

How long is the real wall?

Length: _____

A room is 15 feet long.

How long is it on a drawing with a scale of 1 inch = 3 feet?

Length: _____

A drawing is reduced to $\frac{1}{2}$ of its size.

A 10-cm side becomes how long?

- A.** 2 cm
- B.** 20 cm
- C.** 10 cm
- D.** 5 cm

A drawing that is 4 cm by 6 cm is enlarged by a scale factor of $\frac{3}{2}$.

What are the new dimensions?

Dimensions: _____

True or false?

Using a scale factor less than 1 makes a drawing smaller.

A. True

B. False

Explain the effect of multiplying by a fraction greater than, less than, or equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Fractions equal to 1	3.NF.A.3.c
FORWARD 1	Percents greater than and less than 100%	6.RP.A.3.c
FORWARD 2	Use a multiplier for percent increase and decrease	7.RP.A.3

SET 34 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

True or false?

Multiplying 12 by $\frac{7}{6}$ gives a product greater than 12.

- A.** True
- B.** False

Which statement explains why $\frac{4}{5} \times 30$ is less than 30?

- A.** $\frac{4}{5}$ is less than 1.
- B.** $\frac{4}{5}$ is greater than 1.
- C.** 30 is greater than 1.
- D.** The product is a fraction.

Use the fact that $\frac{4}{4} = 1$ to explain why $\frac{2}{3} = \frac{8}{12}$.

Explanation: _____

Which expression has a value equal to 9?

A. $\frac{6}{5} \times 9$

B. $\frac{5}{6} \times 9$

C. $\frac{1}{5} \times 9$

D. $\frac{5}{5} \times 9$

Without multiplying, order the products from least to greatest.

$$\frac{3}{4} \times 10, \frac{4}{3} \times 10, \frac{4}{4} \times 10$$

Order: _____

SET 34 • 5.NF.B.5.b

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

To write a fraction equal to $\frac{3}{5}$, you can multiply its numerator and denominator by the same number.

- A.** True
- B.** False

Find the missing number.

$$\frac{3}{5} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{7}{8}$?

A. $\frac{14}{24}$

B. $\frac{21}{24}$

C. $\frac{7}{16}$

D. $\frac{8}{7}$

Find the missing number.

$$\frac{1}{3} = \frac{4}{?}$$

Missing number: _____

True or false?

$$\frac{2}{9} = \frac{4}{27}$$

A. True

B. False

SET 34 • 5.NF.B.5.b

BACKWARD 2

Fractions equal to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

True or false?

$$\frac{4}{4} = 4$$

A. True

B. False

Write 1 as a fraction with a denominator of 6.

Fraction: _____

Which fraction is equal to 1?

A. $\frac{0}{9}$

B. $\frac{9}{1}$

C. $\frac{1}{9}$

D. $\frac{9}{9}$

True or false?

$\frac{5}{5}$ is greater than 1.

A. True

B. False

How many eighths make 1 whole?

Eighths: _____

SET 34 • 5.NF.B.5.b

FORWARD 1

Percents greater than and less than 100% • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

True or false?

150% of 40 is less than 40.

A. True

B. False

What is 120% of 50?

Answer: _____

Which value is less than 80?

A. 200% of 80

B. 125% of 80

C. 75% of 80

D. 100% of 80

What is 100% of 37?

Answer: _____

What is 250% of 12?

Answer: _____

SET 34 • 5.NF.B.5.b

FORWARD 2

Use a multiplier for percent increase and decrease • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A \$60 price increases by 15%.

Which expression gives the new price?

A. 1.15×60

B. $60 + 15$

C. 0.85×60

D. 0.15×60

A \$90 jacket is 20% off.

Multiply by a decimal to find the sale price.

Sale price: _____

True or false?

Multiplying a price by 0.7 decreases it by 30%.

A. True

B. False

A population of 2,000 grows by 5%.

What is the new population?

Population: _____

Which multiplier shows a 40% decrease?

A. 0.6

B. 0.4

C. 1.4

D. 1.04

Solve real-world problems by multiplying fractions and mixed numbers

MAIN	Grade 5 main question	5.NF.B.6
BACKWARD 1	Word problems: a whole number times a fraction	4.NF.B.4.c
BACKWARD 2	Mixed numbers as fractions	4.NF.B.3.b
BACKWARD 3	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Word problems with division of fractions	6.NS.A.1
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 35 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is 6 m long. $\frac{2}{3}$ of its length is planted with beans.

How many meters are planted with beans?

Meters: _____

Sam walks $\frac{5}{8}$ mile each day.

How far does he walk in 4 days?

A. $\frac{5}{32}$ mile

B. $\frac{9}{8}$ miles

C. $2\frac{1}{2}$ miles

D. $4\frac{5}{8}$ miles

A board is $3\frac{1}{3}$ feet long. Jo uses $\frac{3}{5}$ of the board.

How many feet does she use?

Feet: _____

A pitcher holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the pitcher?

Liters: _____

SET 35 • 5.NF.B.6

BACKWARD 1

Word problems: a whole number times a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each serving is $\frac{2}{3}$ cup.

How many cups are in 5 servings?

Cups: _____

True or false?

4 children each drink $\frac{3}{8}$ liter of milk. They drink $1\frac{1}{2}$ liters in all.

- A. True
- B. False

Each piece of ribbon is $\frac{5}{6}$ yard long.

How long are 3 pieces in all?

A. $\frac{5}{18}$ yard

B. $2\frac{1}{2}$ yards

C. $3\frac{5}{6}$ yards

D. $\frac{15}{18}$ yard

A bug crawls $\frac{3}{10}$ meter each minute.

How far does it crawl in 7 minutes?

Distance: _____

Each bag holds $\frac{1}{4}$ pound of nuts.

How many pounds are in 6 bags?

Pounds: _____

SET 35 • 5.NF.B.6

BACKWARD 2

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $2\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{1}{2} = \frac{11}{2}$$

A. True

B. False

Which fraction is equal to $4\frac{2}{5}$?

A. $\frac{22}{5}$

B. $\frac{42}{5}$

C. $\frac{11}{5}$

D. $\frac{24}{5}$

Write $\frac{15}{8}$ as a mixed number.

Mixed number: _____

SET 35 • 5.NF.B.6

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{5}{2}$$

Product: _____

Multiply.

$$\frac{2}{3} \times \frac{3}{2}$$

Product: _____

True or false?

$$\frac{10}{3} \times \frac{3}{5} = 2$$

A. True

B. False

$$\frac{3}{2} \times \frac{2}{3} = ?$$

A. $\frac{5}{6}$

B. 1

C. $\frac{6}{5}$

D. $\frac{9}{4}$

Multiply.

$$\frac{5}{8} \times 4$$

Product: _____

SET 35 • 5.NF.B.6

FORWARD 1

Word problems with division of fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{3}{4}$ -cup servings are in 6 cups of soup?

Servings: _____

A $\frac{7}{8}$ -mile trail is split into $\frac{1}{8}$ -mile sections.

How many sections are there?

Sections: _____

A rectangle has an area of $\frac{3}{4}$ square foot and a width of $\frac{1}{2}$ foot.

What is its length?

A. $\frac{3}{8}$ foot

B. $1\frac{1}{4}$ feet

C. $1\frac{1}{2}$ feet

D. $\frac{2}{3}$ foot

Ann has $2\frac{1}{2}$ pounds of clay. Each pot uses $\frac{5}{8}$ pound.

How many pots can she make?

Pots: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = \frac{1}{9}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot per second for $6\frac{1}{2}$ seconds.

What is the diver's change in depth?

Change: _____

The temperature was 2°F . It dropped $\frac{1}{2}^{\circ}\text{F}$ each hour for 9 hours.

What was the final temperature?

Temperature: _____

A stock loses $\frac{3}{8}$ dollar each day for 4 days.

What is the total change in value?

A. $-4\frac{3}{8}$ dollars

B. $-\frac{3}{32}$ dollar

C. $-1\frac{1}{2}$ dollars

D. $1\frac{1}{2}$ dollars

A tank holds $12\frac{1}{2}$ gallons. $\frac{2}{5}$ of the water drains out.

How many gallons drain out?

Gallons: _____

True or false?

$$-\frac{2}{3} \times 4\frac{1}{2} = -3$$

A. True

B. False

Multiply mixed numbers

MAIN

Grade 5 main question

5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1

4.NF.B.3.b

BACKWARD 2

Multiplying fractions

5.NF.B.4.a

BACKWARD 3

Breaking apart a factor to multiply

3.OA.B.5

FORWARD 1

Volume of prisms with fractional edge lengths

6.G.A.2

FORWARD 2

Surface area and volume of prisms

7.G.B.6

SET 36 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

Multiply.

$$2\frac{1}{4} \times 1\frac{2}{3}$$

A. $3\frac{1}{6}$

B. $4\frac{1}{2}$

C. $2\frac{2}{12}$

D. $3\frac{3}{4}$

Multiply.

$$3\frac{1}{5} \times 2\frac{1}{2}$$

Product: _____

True or false?

$$1\frac{1}{2} \times 1\frac{1}{2} = 1\frac{1}{4}$$

A. True

B. False

SET 36 • 5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $1\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{13}{3}$ as a mixed number.

Mixed number: _____

True or false?

$$2\frac{2}{3} = \frac{22}{3}$$

A. True

B. False

Which mixed number is equal to $\frac{9}{2}$?

A. $2\frac{1}{9}$

B. $4\frac{1}{9}$

C. $9\frac{1}{2}$

D. $4\frac{1}{2}$

Write $5\frac{1}{2}$ as a fraction.

Fraction: _____

SET 36 • 5.NF.B.6

BACKWARD 2

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{7}{4} \times \frac{2}{3}$$

Product: _____

True or false?

$$\frac{5}{2} \times \frac{3}{2} = \frac{15}{2}$$

A. True

B. False

$$\frac{4}{3} \times \frac{9}{4} = ?$$

A. 3

B. 2

C. $\frac{36}{7}$

D. $\frac{13}{7}$

Multiply.

$$\frac{5}{3} \times 6$$

Product: _____

Multiply.

$$\frac{3}{2} \times \frac{3}{2}$$

Product: _____

SET 36 • 5.NF.B.6

BACKWARD 3

Breaking apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking
• 3.OA.B.5

True or false?

$$4 \times (5 + 2) = 4 \times 5 + 4 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 12 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 3×15 ?

A. $3 \times 10 + 5$

B. $3 + 10 \times 5$

C. $3 \times 10 \times 5$

D. $3 \times 10 + 3 \times 5$

True or false?

$$7 \times 13 = 7 \times 10 + 3$$

A. True

B. False

Fill in the blank.

$$8 \times (4 + 1) = 8 \times 4 + 8 \times \underline{\quad}$$

Blank: _____

SET 36 • 5.NF.B.6

FORWARD 1

Volume of prisms with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $2\frac{1}{2}$ ft by 2 ft by $1\frac{1}{2}$ ft.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its volume?

Volume: _____

A prism has a base area of $3\frac{1}{3}$ square cm and a height of $1\frac{1}{2}$ cm.

What is its volume?

A. $3\frac{1}{2}$ cubic cm

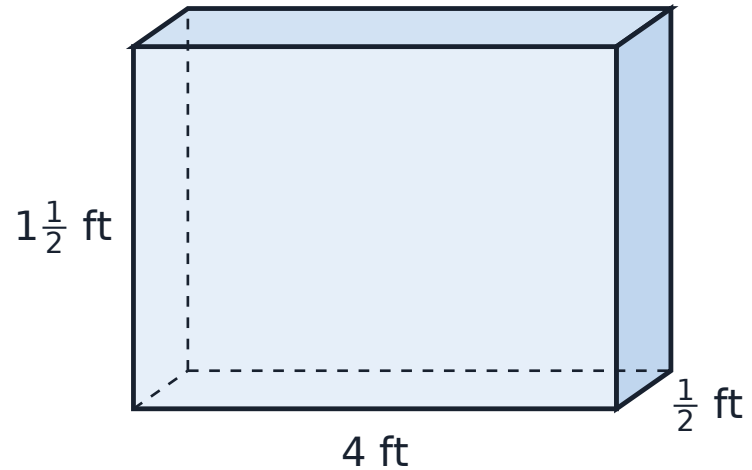
B. 10 cubic cm

C. $4\frac{5}{6}$ cubic cm

D. 5 cubic cm

Find the volume of the prism.

Volume: _____



True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

A. True

B. False

SET 36 • 5.NF.B.6

FORWARD 2

Surface area and volume of prisms • Grade 7 • Geometry • 7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

A triangular prism has triangle bases with a base of 3 in. and a height of $2\frac{1}{2}$ in. The prism is 6 in. long.

What is its volume?

Volume: _____

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its volume?

- A.** 20 cubic inches
- B.** $10\frac{1}{2}$ cubic inches
- C.** 30 cubic inches
- D.** 300 cubic inches

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

True or false?

A prism with a base area of $7\frac{1}{2}$ square feet and a height of 4 feet has a volume of $11\frac{1}{2}$ cubic feet.

A. True

B. False

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 3	Multiplying unit fractions	5.NF.B.4.a
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{3} \div 4$$

Quotient: _____

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

A. True

B. False

Which unit fraction is the smallest?

A. $\frac{1}{8}$

B. $\frac{1}{12}$

C. $\frac{1}{2}$

D. $\frac{1}{5}$

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

A. True

B. False

Which expression shows 21 shared equally among 7 people?

A. $21 \div 7$

B. $21 - 7$

C. $7 \div 21$

D. 21×7

Divide.

$$16 \div 4$$

Quotient: _____

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

SET 37 • 5.NF.B.7.a

BACKWARD 3

Multiplying unit fractions • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{3} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$$

A. True

B. False

$$\frac{1}{6} \times \frac{1}{2} = ?$$

A. $\frac{1}{8}$

B. $\frac{1}{3}$

C. $\frac{2}{6}$

D. $\frac{1}{12}$

Multiply.

$$\frac{1}{4} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{1}{3} = \frac{2}{5}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

True or false?

$$-\left(\frac{1}{2} \div 5\right) = \left(-\frac{1}{2}\right) \div 5$$

A. True

B. False

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$4 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

How many $\frac{1}{3}$ s are in 7?

A. 10

B. 21

C. $\frac{7}{3}$

D. $\frac{1}{21}$

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

A. True

B. False

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

How many 4s are in 32?

Answer: _____

A 30-inch ribbon is cut into 5-inch pieces.
How many pieces are there?

Pieces: _____

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

True or false?

There are 3 thirds in 2 wholes.

A. True

B. False

How many $\frac{1}{5}$ s make 2 wholes?

- A.** 2
- B.** 7
- C.** 10
- D.** 5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

There are 2 halves in 2 wholes.

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

How many $\frac{2}{3}$ -cup servings are in 8 cups?

A. 6

B. $\frac{1}{12}$

C. $\frac{16}{3}$

D. 12

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A.** $\frac{1}{12}$ mile per hour
- B.** 3 miles per hour
- C.** 8 miles per hour
- D.** 12 miles per hour

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A.** True
- B.** False

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	A fraction as division	5.NF.B.3
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{3}$ -cup servings of raisins are in 2 cups?

Servings: _____

3 friends share $\frac{1}{2}$ pound of chocolate equally.

How much chocolate does each friend get?

Pounds: _____

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

A. True

B. False

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.
How many bracelets can she make?

Bracelets: _____

True or false?

If 32 students sit 4 at each table, there are 6 tables.

A. True

B. False

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A.** 8
- B.** 200
- C.** 35
- D.** 45

A 27-foot fence is split into 9 equal sections.
How long is each section?

Length: _____

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

SET 39 • 5.NF.B.7.c

BACKWARD 2

A fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $1 \div 4$ as a fraction.

Fraction: _____

True or false?

$$2 \div 3 = \frac{2}{3}$$

A. True

B. False

4 people share 1 pie equally.

How much pie does each person get?

A. $\frac{1}{4}$ pie

B. $\frac{4}{1}$ pies

C. $\frac{3}{4}$ pie

D. 4 pies

Write $\frac{1}{6}$ as a division expression.

Expression: _____

True or false?

5 people share 2 sandwiches equally, so each person gets $\frac{5}{2}$ sandwiches.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

A. $\$135$

B. $-\$45$

C. $-\$15$

D. $-\$135$

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

A. True

B. False

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 3	Multiplying decimals by powers of 10	5.NBT.A.2
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

How many ounces are in 3 pounds?

Ounces: _____

A track is 2.5 kilometers long.

How many meters is that?

- A.** 250 m
- B.** 25,000 m
- C.** 2,500 m
- D.** 25 m

Convert.

6 yards = ____ feet

Feet: _____

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

True or false?

1 pound = 16 ounces

A. True

B. False

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

How many meters are in 1 kilometer?

Meters: _____

True or false?

1 yard = 12 inches

A. True

B. False

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$16 \times 3$$

Product: _____

Multiply.

$$1,000 \times 7$$

Product: _____

True or false?

$$3 \times 36 = 98$$

A. True

B. False

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

Multiply.

$$60 \times 5$$

Product: _____

SET 40 • 5.MD.A.1

BACKWARD 3

Multiplying decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$2.5 \times 1,000$$

Product: _____

True or false?

$$0.75 \times 100 = 7.5$$

A. True

B. False

$$4.2 \times 10 = ?$$

A. 0.42

B. 420

C. 42

D. 4.12

Multiply.

$$1.6 \times 1,000$$

Product: _____

Multiply.

$$0.3 \times 100$$

Product: _____

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.

How many centimeters are in 5 inches?

Centimeters: _____

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A. True
- B. False

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A. True
- B. False

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Dividing by powers of 10	5.NBT.A.2
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

30 inches = ____ feet

Feet: _____

Convert.

750 milliliters = ____ liters

Liters: _____

A rope is 45 feet long.

How many yards is that?

A. 15

B. 5

C. 135

D. 42

Convert.

40 ounces = ____ pounds

Pounds: _____

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

True or false?

1 kilogram = 1,000 grams

A. True

B. False

How many feet are in 1 yard?

A. 10

B. 3

C. 36

D. 12

How many cups are in 1 pint?

Cups: _____

True or false?

1 foot = 10 inches

A. True

B. False

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$36 \div 3$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$84 \div 7 = 14$$

A. True

B. False

$$90 \div 3 = ?$$

A. 33

B. 300

C. 27

D. 30

Divide.

$$5,000 \div 5$$

Quotient: _____

SET 41 • 5.MD.A.1

BACKWARD 3

Dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Divide.

$$750 \div 1,000$$

Quotient: _____

True or false?

$$1,350 \div 1,000 = 13.5$$

A. True

B. False

$$45 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 0.045
- D.** 4,500

Divide.

$$2,600 \div 1,000$$

Quotient: _____

Divide.

$$80 \div 100$$

Quotient: _____

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

Write an equation for the number of feet f in y yards.

Equation: _____

Which equation converts k kilograms to g grams?

A. $k = 1,000g$

B. $g = 1,000k$

C. $g = k \div 1,000$

D. $g = k + 1,000$

Write an equation for the number of pounds p in z ounces.

Equation: _____

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A. True
- B. False

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.
How many feet of ribbon are left?

Feet: _____

A recipe needs 2 pounds of flour. Jo has 20 ounces.
How many more ounces does she need?

Ounces: _____

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A.** 1,050 m
- B.** 2,050 m
- C.** 9,700 m
- D.** 862 m

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.
What time does it end?

Time: _____

True or false?

3 quarters and 2 dimes make 85 cents.

A. True

B. False

Jake has 2 meters of string. He uses 75 cm.

How many centimeters are left?

A. 1.25 cm

B. 75 cm

C. 125 cm

D. 275 cm

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

True or false?

1 gallon = 4 quarts

A. True

B. False

How many centimeters are in 1 meter?

- A.** 1,000
- B.** 100
- C.** 10
- D.** 12

How many feet are in 3 yards?

Feet: _____

True or false?

1 liter = 100 milliliters

A. True

B. False

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

A. \$72

B. \$18

C. \$24

D. \$2

A 24-ounce jar of juice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A. True
- B. False

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

A. 1,050

B. 3

C. 4

D. 5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

A. True

B. False

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare two distributions shown in dot plots	7.SP.B.3

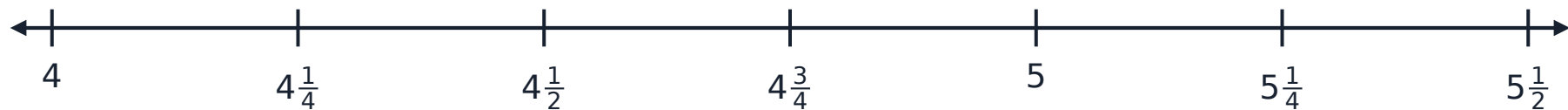
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

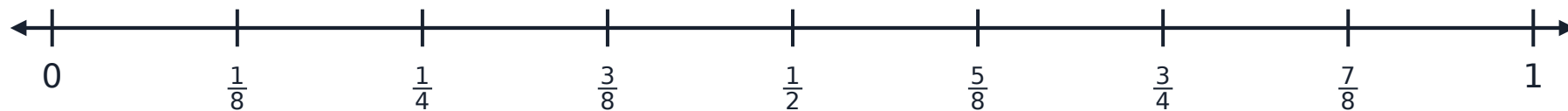
Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



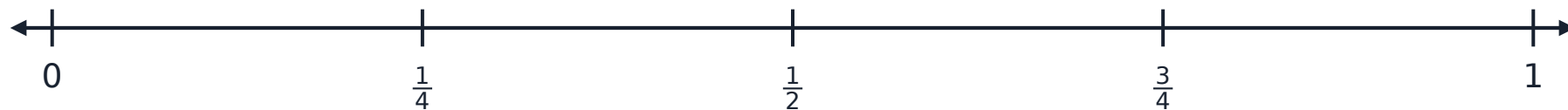
Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1 , $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



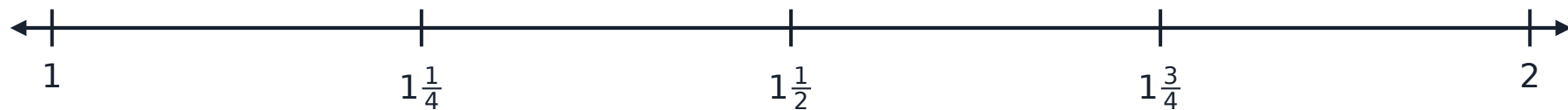
Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



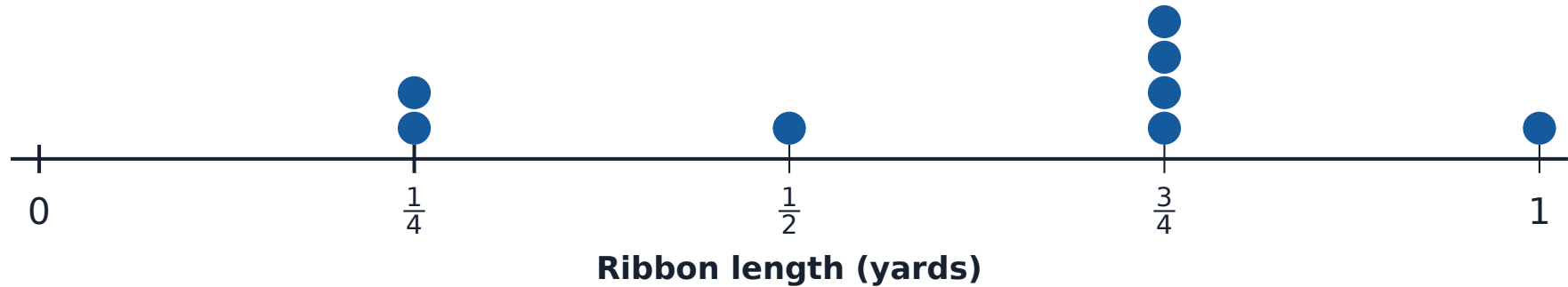
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

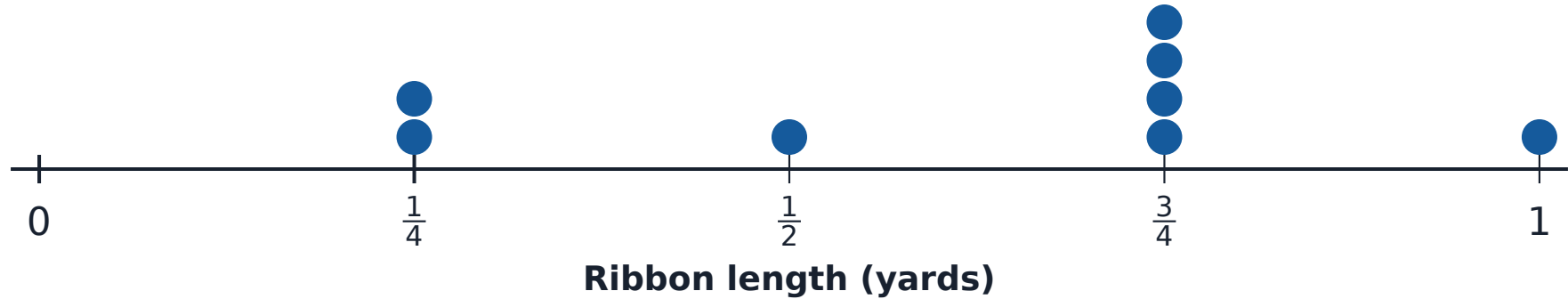
How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

True or false?

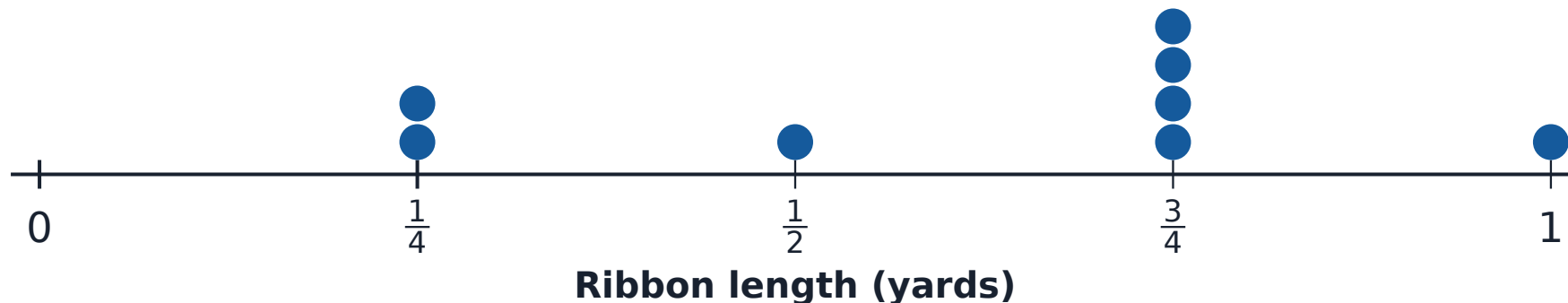
Two ribbons are $\frac{1}{4}$ yard long.



A. True

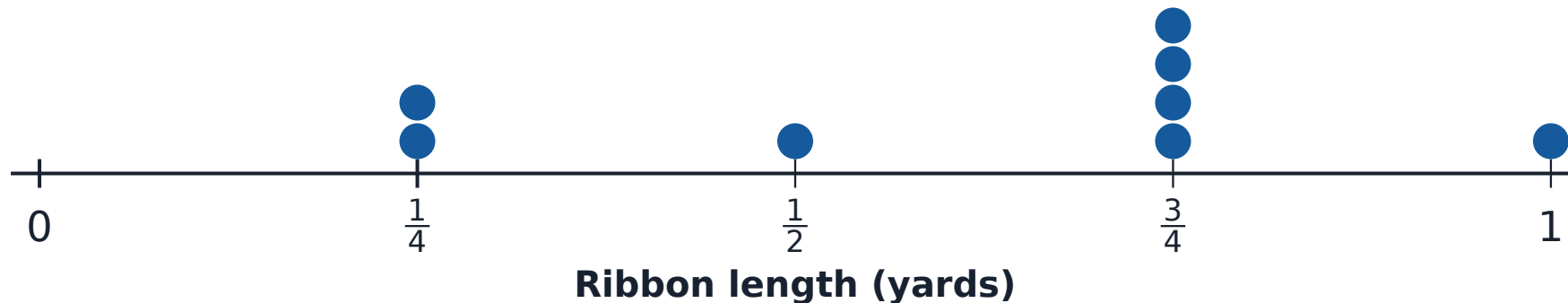
B. False

Which ribbon length is the most common?



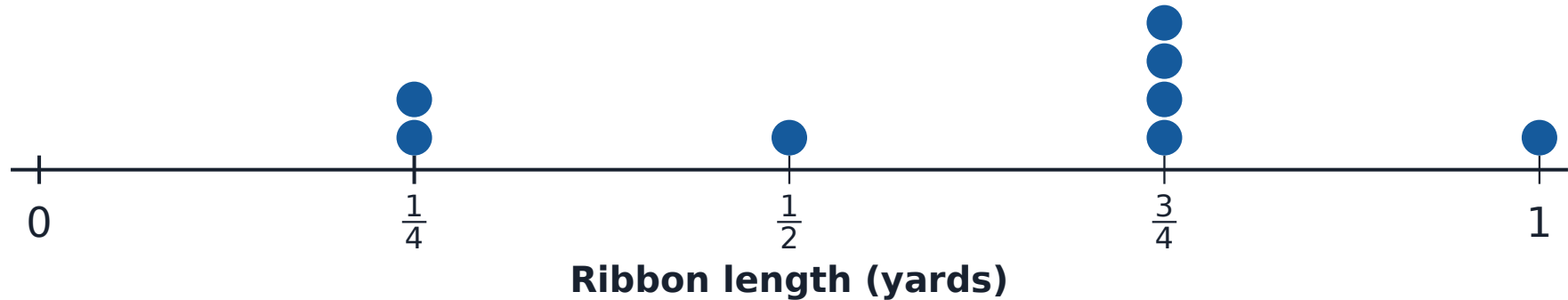
- A.** 1 yard **B.** $\frac{1}{2}$ yard **C.** $\frac{1}{4}$ yard **D.** $\frac{3}{4}$ yard

How many ribbons were measured?



Ribbons: _____

What is the difference between the longest and the shortest ribbon?



Difference: _____

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



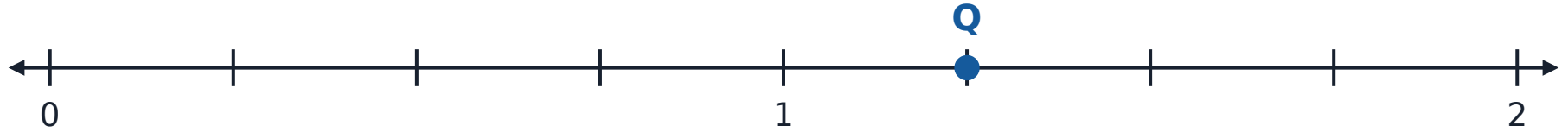
A. 5

B. 7

C. 6

D. 1

What number is at point Q?



Q = _____

True or false?

On a number line from 0 to 1 marked in eighths, $\frac{6}{8}$ is one mark to the left of 1.

- A. True
- B. False

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line would you use for a line plot of 3, 4, 4, 7?

- A.** A number line from 0 to 8
- B.** A number line from 0 to 3
- C.** A number line from 5 to 6
- D.** A number line from 10 to 20

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

True or false?

A dot plot shows every value in a data set.

- A.** True
- B.** False

SET 43 • 5.MD.B.2

FORWARD 2

Compare two distributions shown in dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

About how many inches greater is the center of Plot B than the center of Plot A?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Inches: _____

True or false?

The two plots have about the same spread.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Which statement is true?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

- A.** Plants in Plot A are typically taller. **B.** Plot A has much more spread.
C. The centers are equal. **D.** Plants in Plot B are typically taller.

What is the range of the heights in Plot A?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Range: _____

True or false?

The two dot plots overlap.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Use operations on fractions to solve problems with line plot data

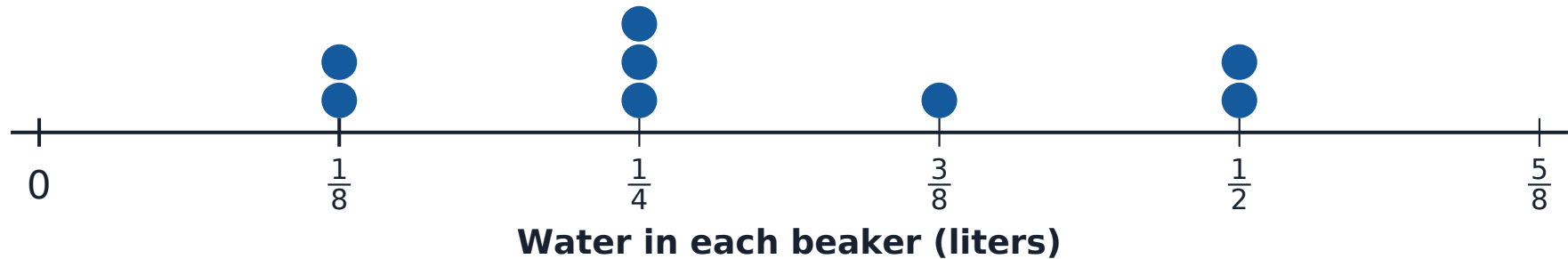
MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 2	Adding and subtracting fractions with unlike denominators	5.NF.A.1
BACKWARD 3	Sharing a total equally	5.NF.B.3
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.
What is the total amount of water?



Total: _____

If all of the water were shared equally among the beakers, how much would each beaker hold?



Liters: _____

How much more water is in the fullest beaker than in the least full beaker?



- A.** $\frac{1}{2}$ liter **B.** $\frac{1}{8}$ liter **C.** $\frac{5}{8}$ liter **D.** $\frac{3}{8}$ liter

What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.



A. True

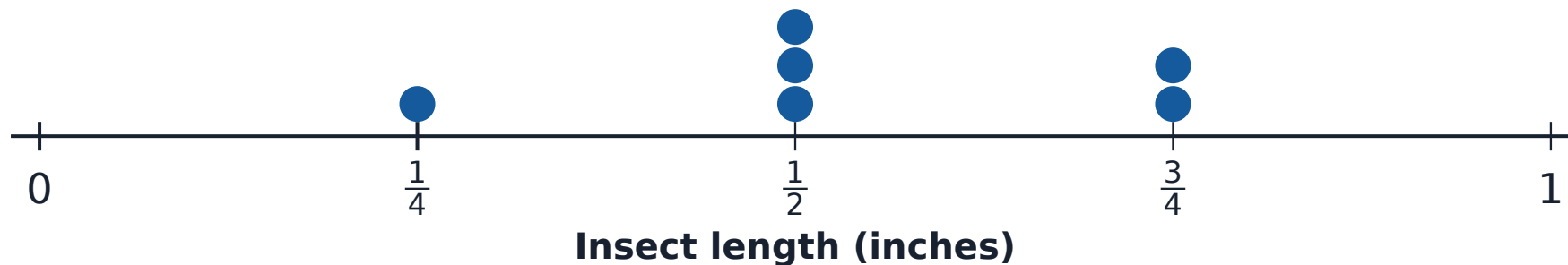
B. False

SET 44 • 5.MD.B.2

BACKWARD 1

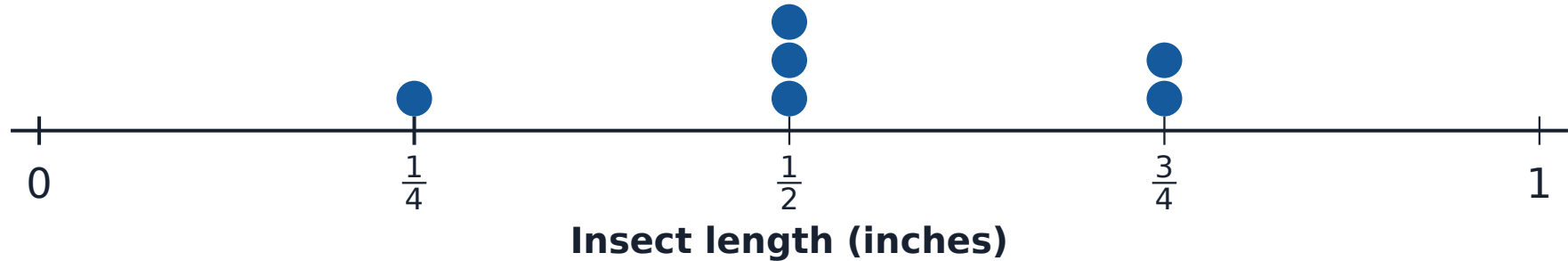
Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

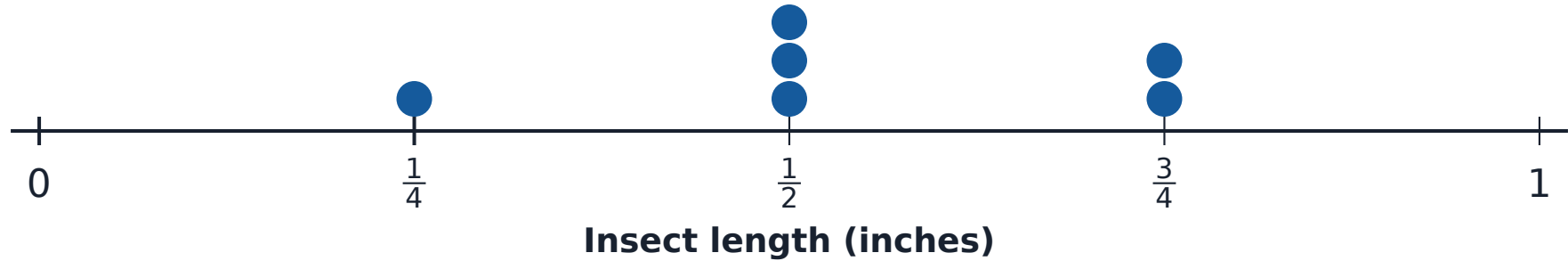
What is the difference between the longest and the shortest insect?



Difference: _____

True or false?

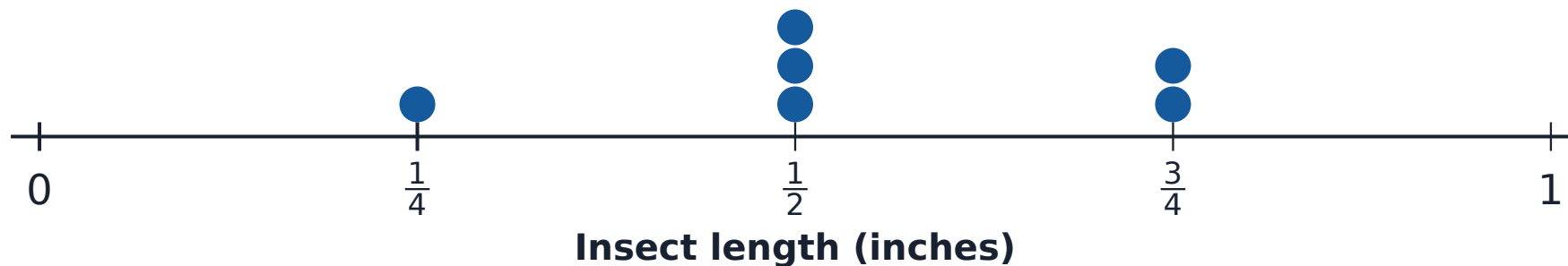
The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.



A. True

B. False

How many insects were measured?



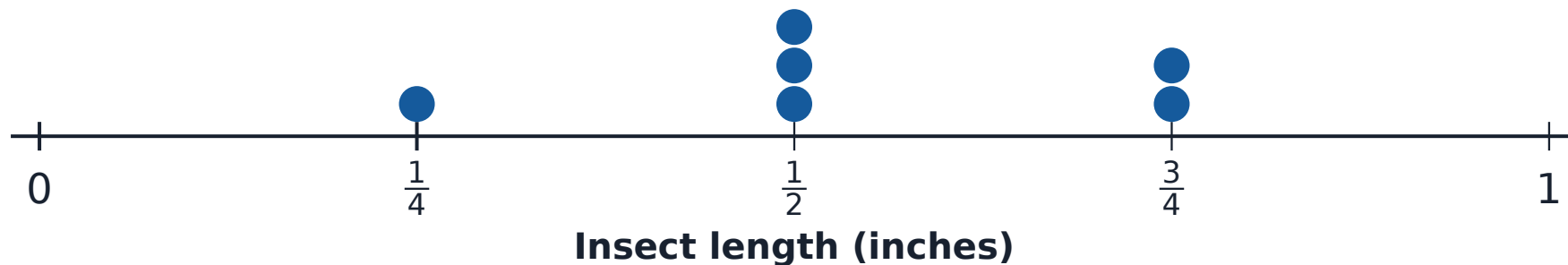
A. 4

B. 3

C. 5

D. 6

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

SET 44 • 5.MD.B.2

BACKWARD 2

Adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{8} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{8} + \frac{1}{2}$$

Sum: _____

True or false?

$$\frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{16}$$

A. True

B. False

$$\frac{1}{2} + \frac{3}{8} + \frac{1}{4} = ?$$

A. $1\frac{3}{8}$

B. $\frac{7}{8}$

C. $1\frac{1}{8}$

D. $\frac{5}{14}$

Subtract.

$$\frac{1}{2} - \frac{1}{8}$$

Difference: _____

SET 44 • 5.MD.B.2

BACKWARD 3

Sharing a total equally • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

3 liters of water are shared equally among 8 beakers.
How much water does each beaker get?

Liters: _____

True or false?

5 cups shared equally among 4 jars gives $\frac{4}{5}$ cup per jar.

- A.** True
- B.** False

2 pounds of clay are shared equally by 6 students.

How much clay does each student get?

A. $\frac{1}{3}$ pound

B. $\frac{1}{6}$ pound

C. 3 pounds

D. $\frac{2}{3}$ pound

7 meters of rope are cut into 4 equal pieces.

How long is each piece?

Length: _____

Write $9 \div 4$ as a mixed number.

Mixed number: _____

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

Find the mean.

12, 15, 21

Mean: _____

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

The dot plot shows the goals scored in 6 games.
What is the mean number of goals?



Mean: _____

True or false?

The mean of 4, 4, and 10 is 4.

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Plants in the second population tend to be taller.
- C.** Plants in the first population tend to be taller.
- D.** No conclusion is possible.

True or false?

Comparing the means of random samples is one way to compare two populations.

A. True

B. False

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

Sample A has a mean of 40 and a MAD of 5. Sample B has a mean of 50 and a MAD of 5.

The difference in means is how many times the MAD?

Times the MAD: _____

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

True or false?

Volume measures the amount of space inside a solid figure.

- A.** True
- B.** False

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

A. True

B. False

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

How many unit squares cover a 3-by-3 square?

Unit squares: _____

True or false?

Area is measured in cubic units.

A. True

B. False

A square has sides that are 1 inch long.

What is its area?

Area: _____

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

True or false?

All the faces of a cube are squares.

A. True

B. False

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

How many vertices (corners) does a cube have?

Vertices: _____

True or false?

All the edges of a cube are the same length.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

A. $\frac{1}{3}$ cubic foot

B. $\frac{1}{9}$ cubic foot

C. $\frac{1}{27}$ cubic foot

D. 1 cubic foot

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

- A. True
- B. False

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

A. 60 cubic inches

B. 30 cubic inches

C. 16 cubic inches

D. 600 cubic inches

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

True or false?

Volume is measured in cubic units.

A. True

B. False

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

- A. True
- B. False

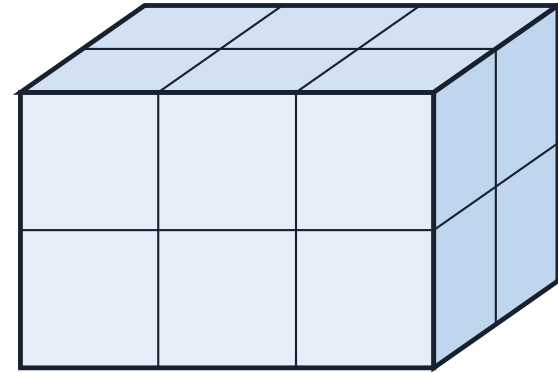
A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units
- B.** 18 cubic units
- C.** 54 cubic units
- D.** 6 cubic units

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

Overlapping squares give a wrong count for area.

A. True

B. False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 40 square units
- C.** 20 cubic units
- D.** 20 square units

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

True or false?

Area can be found by counting the unit squares that cover a shape.

- A.** True
- B.** False

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

True or false?

3 rows of 6 make 9.

A. True

B. False

Which equation matches an array with 2 rows of 7?

A. $2 + 7 = 9$

B. $7 + 7 = 14$

C. $7 - 2 = 5$

D. $2 + 2 = 4$

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

Write an addition equation for 3 rows of 4.

Equation: _____

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

A. 27 cubic feet

B. 9 cubic feet

C. 1 cubic foot

D. 3 cubic feet

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A. True
- B. False

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6